XNO:3
.2-2026

DETERMINATION OF VOLTAGE IN CIRCUIT USING NODAL ANALYSIS

AIM:

To determine the Voltage in the circuit using nodal analysis both theoretically and pratically for a given DC Circuit.

APPARATUS REQUIRED:

S.NO	APPARATUS	SPECIFICATION	QUANTITY
1)	Regulated Power Supply (RPS)	(0-30 V)	1
2)	Multimeter	-	1
3)	Resistor	470Ω, 330Ω	3, 2
4)	Bread Board	-	1

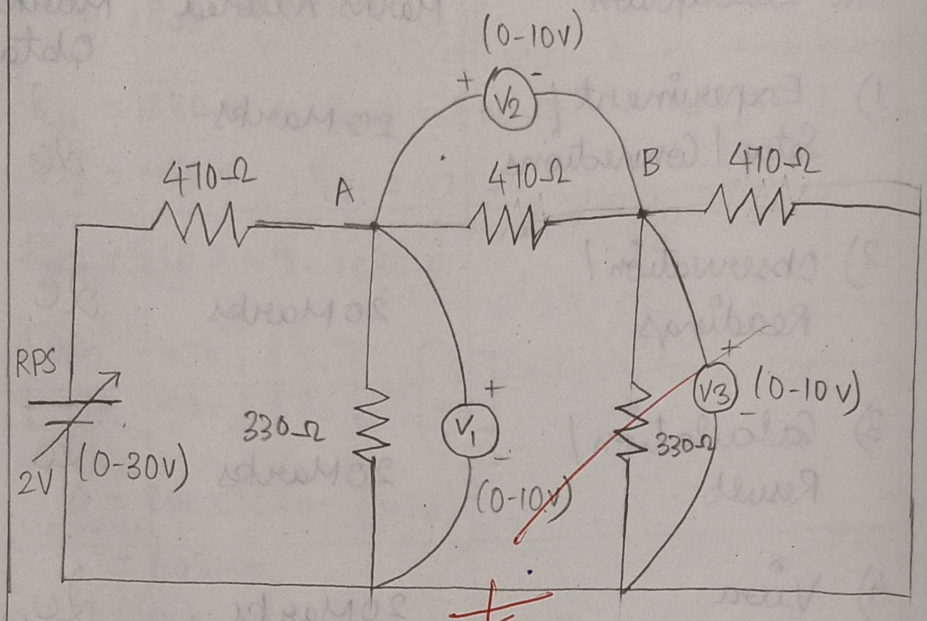
PROCEDURE:

1) Give connections as per the circuit diagram

2) Switch ON the supply, Vary the RPS (Regulated Power Supply) and set a particular input voltage.

3) Note down the readings of ammeter and voltmeter and tabulate them.

CIRCUIT DIAGRAM:



Nodal Circuit

4) Vary the RPS for different input voltages and note down the readings of all the meters.

5) Reduce the RPS to its minimum value and Switch OFF the Supply.

6) Using the tabulated values, Verify Kirchoff's laws pratically and Verify it theoretically.

TABULAR COLUMN:

Parameters	Theoretical	Pratical
Supply Voltage	2V	2.V
V_A	0.64V	0.49
V_B	0.19V	0.35
V_{AB}	0.45V	0.14

O/P verified

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CALCULATION :

Node A :

$$\frac{V_A - 2}{470} + \frac{V_A}{330} + \frac{V_A - V_B}{470} = 0$$

$$113V_A + 33V_B = 66 \rightarrow (1)$$

Node B :

$$\frac{V_B - V_A}{470} + \frac{V_B}{330} + \frac{V_B}{470} = 0$$

$$113V_A - 33V_B = 0 \rightarrow (2)$$

From (1) & (2)

$V_B = 0.19V$

$V_A = 0.64V$

MARKS SPILT UP:-

S.NO	DESCRIPTION	MARKS ALLOTED	MARKS OBTAINED
1)	Experiment / Setup / Connections	20 Marks	20
2)	Observation / Readings	20 Marks	20
3)	Calculation / Result	20 Marks	20
4)	Viva	20 Marks	20
5)	Record	20 Marks	20
		Total	100

100

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RESULT:

Thus, The Nodal analysis is verified pratically and theoretically. The resultant voltages for 2 V Supply are:

a) The Voltage V_1 is 0.49 V

b) The Voltage V_2 is 0.35 V

c) The Voltage V_3 is 0.14 V